Dashboard Mode

- Real Data — VitalDB
- Synthetic — Deep Dive

Patient Simulator

View Mode

- Single Patient
- Side-by-Side
- CA vs No-CA Overlay

Noise level

1.00

Patient A

Seed A

42

Duration A (min)

480

CA Onset A (min)

380

HR Rise Rate A

0.12

BP Drop Rate A

0.15

SpO2 Drop Rate A

0.06

Baseline HR A

INTRAOPERATIVE CARDIAC ARREST RISK MONITOR

SYNTHETIC DEEP DIVE

Configurable Patient Simulator · All waveforms algorithmically generated | 12 Apr 2026 · 14:15 | CA vs No-CA Overlay

Sachu Mon P. Sajeev · MSc Data Science & AI · TSI University Riga · 2026

TAN · AUROC	LightGBM · AUROC	TAN · AUPRC
0.9937	0.9073	0.6027
95% CI [0.9909–0.9961]	60-min window	240-min window

Synthetic Mode — All waveforms are algorithmically generated from configurable parameters. Adjust deterioration rates, onset timing, and noise in the sidebar. Perfect for thesis presentation and clinical model explanation.

RISK COMPARISON SUMMARY

Patient A — ⚠ CA EVENT

52.4%

HIGH

Patient B — ✓ NO CA

4.1%

LOW

SIGNAL OVERLAY — CA VS NO-CA

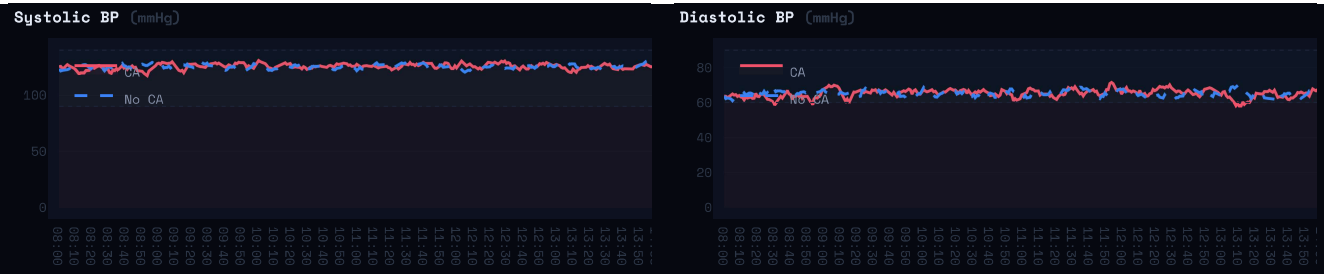
Red = CA patient · Blue dashed = No CA · Shaded = normal range

Heart Rate (bpm)

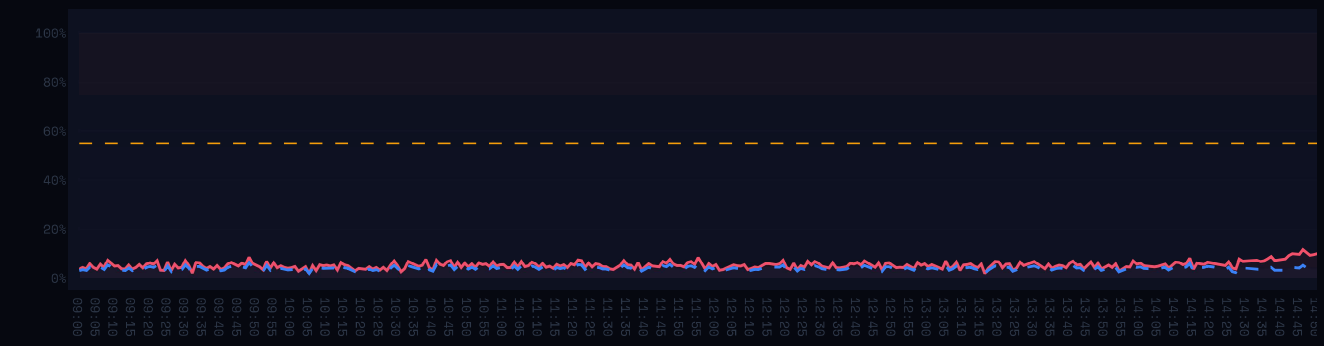
SpO₂ (%)

EtCO₂ (mmHg)

Mean BP (mmHg)



RISK SCORE OVERLAY



Patient #42 ⚠ CA EVENT · SYNTHETIC · Onset: 380min

CA RISK · 60MIN WINDOW	HEART RATE	SPO₂	ETCO₂	MEAN BP	SYSTOLIC BP	DIASTOLIC BP
52.4%	86	95	34	69	110	49
HIGH	bpm	%	mmHg	mmHg	mmHg	mmHg
	→ OK	↓ LOW	↓ LOW	↓ LOW	→ OK	↓ LOW
	→0.2/30m	→0.8/30m	↘2.0/30m	↘3.3/30m	↘3.0/30m	↘2.9/30m
	Range: 69–86	Range: 94–98	Range: 33–40	Range: 68–90	Range: 109–130	Range: 48–71
	Normal: 50–100	Normal: 95–100	Normal: 35–45	Normal: 70–105	Normal: 90–140	Normal: 60–90

DETERIORATION PHASE TIMELINE

STABLE PHASE	EARLY DETERIORATION	ACTIVE CRISIS
380min	40min	41min
0–380min mark	380–420min mark	420–461min mark

RISK SCORE OVER TIME

VITAL SIGNAL TRENDS

Waveform algorithmically generated from configurable parameters.

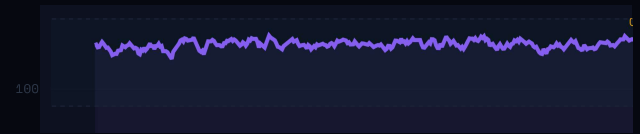
Heart Rate (bpm) · normal 50–100

SpO₂ (%) · normal 95–100

EtCO₂ (mmHg) · normal 35–45

Mean BP (mmHg) · normal 70–105

Systolic BP (mmHg) · normal 90–140

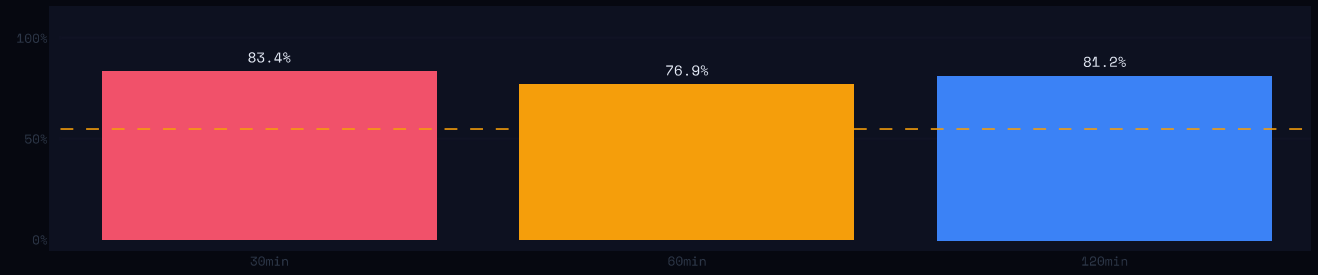


Diastolic BP (mmHg) · normal 60–90

CHANGE POINT LOG

No significant change points detected.

RISK ACROSS AVAILABLE WINDOWS



Patient #77 ✓ NO CA · SYNTHETIC

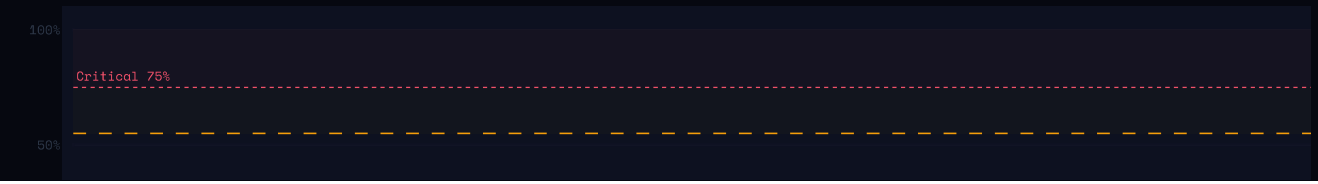
CA RISK · 60MIN WINDOW

4.1 %

LOW

HEART RATE	SPO₂	ETCO₂	MEAN BP	SYSTOLIC BP	DIASTOLIC BP
71	98	38	85	127	66
bpm	%	mmHg	mmHg	mmHg	mmHg
→ OK	→ OK	→ OK	→ OK	→ OK	→ OK
→0.1/30m	→0.0/30m	→0.1/30m	→0.7/30m	→0.3/30m	→0.2/30m
Range: 66–78	Range: 97–98	Range: 36–39	Range: 82–88	Range: 120–131	Range: 61–71
Normal: 50–100	Normal: 95–100	Normal: 35–45	Normal: 70–105	Normal: 90–140	Normal: 60–90

RISK SCORE OVER TIME



VITAL SIGNAL TRENDS

Waveform algorithmically generated from configurable parameters.

Heart Rate (bpm) · normal 50–100

SpO_2 (%) · normal 95–100

$EtCO_2$ (mmHg) · normal 35–45

Mean BP (mmHg) · normal 70–105

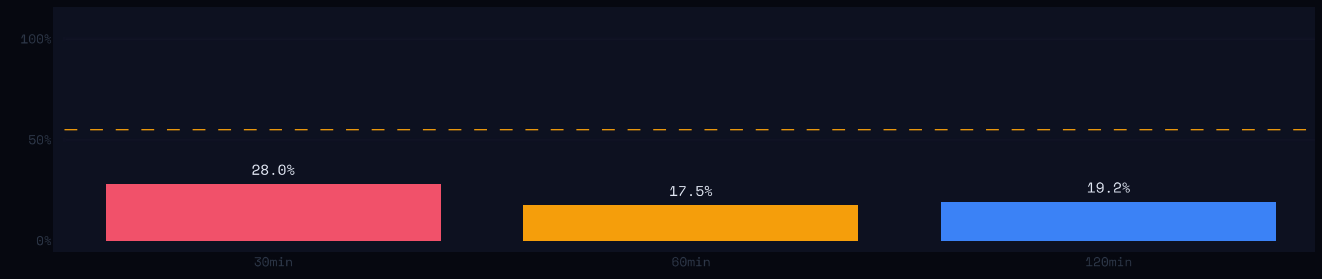
Systolic BP (mmHg) · normal 90–140

Diastolic BP (mmHg) · normal 60–90

CHANGE POINT LOG

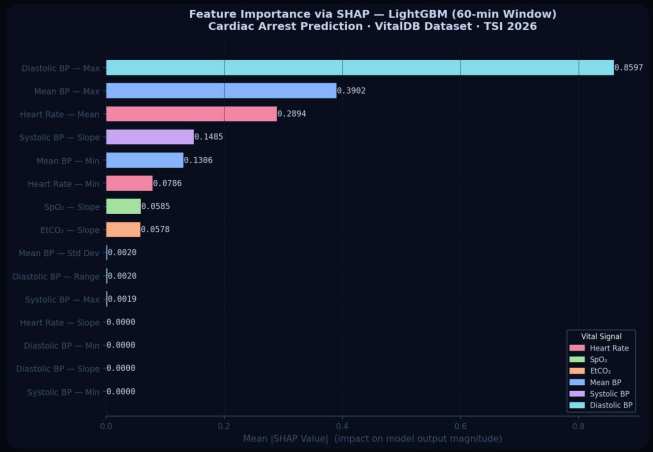
No significant change points detected.

RISK ACROSS AVAILABLE WINDOWS

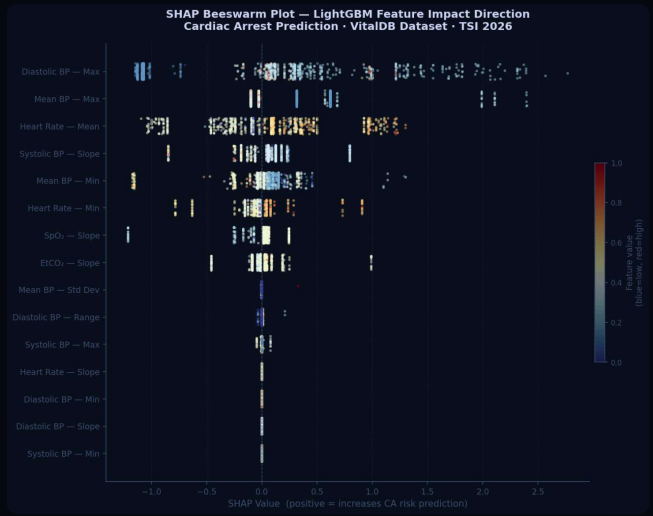


🔬 SHAP Feature Interpretability — LightGBM (60-min Window)

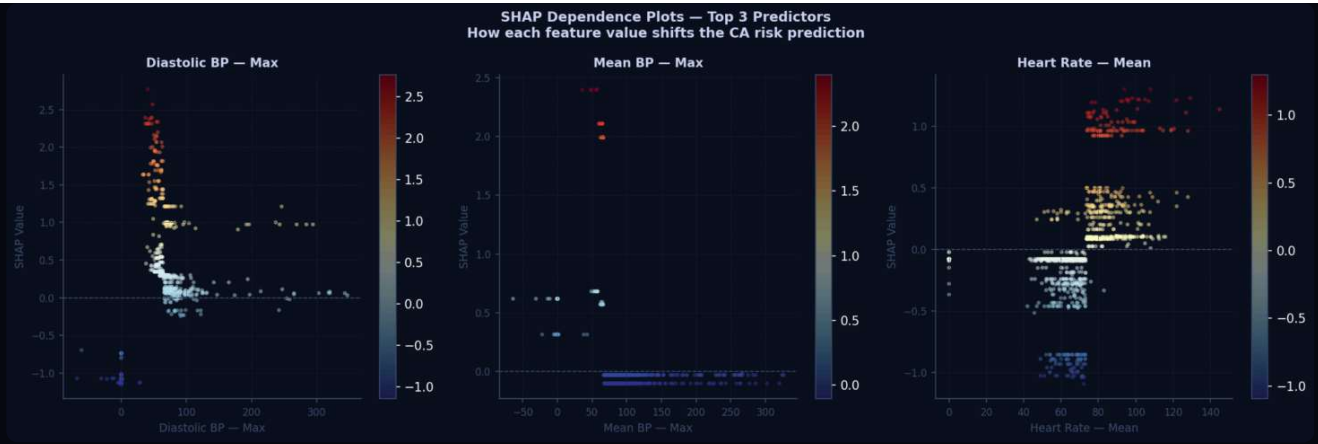
SHapley Additive exPlanations · TreeExplainer · AUROC 0.9073 · VitalDB Dataset · TSI 2026



Mean |SHAP| — Top 15 Features



SHAP Beeswarm — Feature Impact Direction



SHAP Dependence — Top 3 predictors

🔬 **How to read these charts:**

Bar chart — average magnitude of each feature's impact. Longer bar = stronger predictor.

Beeswarm — each dot is one patient. Red dots pushed right = high value increases CA risk. Blue pushed left = decreases risk.

Key finding — Diastolic BP Max, Mean BP Max, and Heart Rate Mean are the top 3 predictors. Run `python shap_analysis.py` to regenerate.

🟢 **240min** — Full 4-hour window | 🔵 **120min** — 2-hour window | 🟡 **60min** — 1-hour window | 🔴 **30min** — Minimal 30-min window

🔴 CRITICAL ≥75% — Alert physician immediately | 🟠 HIGH 50–75% — Increase monitoring | 🟡 MODERATE 25–50% — Watch closely | 🔵 LOW <25% — Routine monitoring